1. Learning guarantees

1. Population error

Let $\mathcal{D}$ be a distribution on inputs, f the binary labeling function, and $\mathcal{H}$ the target class. The setting is **realizable** if a member of $\mathcal{H}$ agrees with f with probability one. The population error of a classifier h is

$$R_{\mathcal{D},f}(h) = \Pr_{X\sim\mathcal{D}}(h(X) \neq f(X)).$$

This probability concerns one new input. Confidence in a learned classifier concerns the random draw of the training sample S and any randomness of the learner A.

2. Weak and strong PAC learning

Weak. Fix an edge $\gamma \in (0, \frac{1}{2})$. One learner and one sample-size function work for every confidence parameter $\delta \in (0,1)$ and every realizable $(\mathcal{D}, f)$. For at least $m_{\mathcal{H}}(\delta)$ independent examples,

$$\Pr\left(R_{\mathcal{D},f}(A(S)) \leq \frac{1}{2} - \gamma\right) \geq 1 - \delta.$$

Strong. For every requested accuracy $\varepsilon \in (0,1)$, replace $\frac{1}{2} - \gamma$ by ε and the sample requirement by $m_{\mathcal{H}}(\varepsilon, \delta)$. The sample requirement in either definition is independent of the unknown distribution. Computational efficiency requires a separate resource bound.

2. Scores and training weights

1. Classifier votes

Fix m labeled rows (x_i, y_i), indexed by i. Each label y_i and base prediction $h_t(x_i)$ is $+1$ or -1. In round t, coefficient α_t weights classifier h_t. After t rounds the vote has score

$$F_t(x) = \sum_{r=1}^{t} \alpha_r h_r(x), \quad F_0(x) = 0.$$

Predict $+1$ if $F_t(x) \geq 0$, and -1 otherwise. The **signed score** $s_i = y_i F_t(x_i)$ is positive for a correct prediction away from ties. At a tie, only a negative label is misclassified.

2. From a score to row weights

The exponential loss of a row with signed score s is $\exp(-s)$. The mean training loss of a score F is

$$L(F) = \frac{1}{m}\sum_{i=1}^{m} \exp(-y_i F(x_i)).$$

Write $L_t = L(F_t)$. Before fitting h_t, take the row losses of the entire current vote F_{t-1} and normalize:

$$w_i = \exp(-y_i F_{t-1}(x_i)),$$

$$D_t(i) = \frac{w_i}{\sum_{j=1}^{m} w_j} = \frac{w_i}{mL_{t-1}}.$$

Smaller signed scores receive greater mass, even on correctly classified rows. Start with $L_0 = 1$ and $D_1(i) = \frac{1}{m}$. The masses sum to one over the fixed training rows.

3. Fit and weight one classifier

1. Check the weighted error first

Fit h_t using D_t. Sum the masses of the rows it gets wrong and compare the result with one half:

$$\varepsilon_t = \sum_{i:h_t(x_i)\neq y_i} D_t(i), \quad \gamma_t = \frac{1}{2} - \varepsilon_t.$$

Here γ_t is the **observed round edge**. A weak PAC guarantee applies across population distributions.

Error	Action
0	All row masses are positive. Stop and return h_t, which is correct on every training row.
$(0, \frac{1}{2})$	Take the finite positive step below.
$\frac{1}{2}$	Zero step. Stop or request another classifier.
$> \frac{1}{2}$	Reject or stop. Flip predictions only if $-h_t$ is allowed, then recheck its error.

2. Choose the vote coefficient

For fixed $F = F_{t-1}$ and $h = h_t$, adding weight a multiplies the loss by

$$g(a) = \frac{L(F + ah)}{L(F)} = (1 - \varepsilon_t)e^{-a} + \varepsilon_t e^{a}.$$

At the minimum, $(1 - \varepsilon_t)e^{-a} = \varepsilon_t e^{a}$. The positive second derivative gives a unique finite step for $0 < \varepsilon_t < \frac{1}{2}$:

$$\alpha_t = \frac{1}{2}\log\frac{1 - \varepsilon_t}{\varepsilon_t}, \quad F_t = F_{t-1} + \alpha_t h_t.$$

A smaller weighted error gives a larger vote.

4. Reweight the same rows

1. Normalize after adding the vote

A row correct under h_t has $y_i h_t(x_i) = +1$ and multiplier $e^{-\alpha_t}$. A wrong row has multiplier e^{α_t}. Divide by the sum of the updated masses:

$$Z_t = \sum_{j=1}^{m} D_t(j) e^{-\alpha_t y_j h_t(x_j)},$$

$$D_{t+1}(i) = \frac{D_t(i) e^{-\alpha_t y_i h_t(x_i)}}{Z_t}.$$

With the exact coefficient, each group now carries half the mass. Within each group, rows keep their relative weights. Use D_{t+1} for the next fit.

2. Two rounds on four fixed rows

Labels are $(+1, +1, -1, -1)$. Classifier h_1 misses only row 3 and h_2 misses only row 1.

Quantity	Round 1	Round 2
Error ε_t	$\frac{1}{4}$	$\frac{1}{6}$
Vote α_t	$\frac{\log 3}{2}$	$\frac{\log 5}{2}$
Factor Z_t	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{5}}{3}$

$$D_2 = \left(\frac{1}{6}, \frac{1}{6}, \frac{1}{2}, \frac{1}{6}\right),$$

$$D_3 = \left(\frac{1}{2}, \frac{1}{10}, \frac{3}{10}, \frac{1}{10}\right).$$

Both classifiers miss one row. The second misses less weight, so its vote is larger. Row 3 becomes correct and row 1 becomes wrong. Training error stays at $\frac{1}{4}$ while the mean loss falls to $\frac{\sqrt{15}}{6}$.

5. Bound the training error

1. Why the factors multiply

Let $\hat{y}_T(x)$ be the label predicted by F_T after T finite rounds. The indicator $\mathbb{1}[E]$ is one when E holds. For signed score $s_i = y_i F_T(x_i)$,

$$\mathbb{1}[\hat{y}_T(x_i) \neq y_i] \leq \mathbb{1}[s_i \leq 0] \leq e^{-s_i}.$$

Averaging bounds the fraction of training mistakes $\hat{R}_S$ by L_T. In each round, $L_t = L_{t-1} Z_t$ and $L_0 = 1$, so

$$\hat{R}_S \leq L_T = \prod_{t=1}^{T} Z_t.$$

The product equals the mean loss. The observed mistake fraction may be smaller.

2. Varying and common edges

Assume every executed round has $0 < \varepsilon_t < \frac{1}{2}$ and uses the exact update. Substitution gives

$$Z_t = 2\sqrt{\varepsilon_t(1-\varepsilon_t)} = \sqrt{1 - 4\gamma_t^2}.$$

Using $1 - u \leq e^{-u}$ gives

$$\hat{R}_S \leq \prod_{t=1}^{T} Z_t \leq \exp\left(-2 \sum_{t=1}^{T} \gamma_t^2\right).$$

If **every** $\gamma_t \geq \gamma > 0$, a common bound is $\exp(-2T\gamma^2)$. For randomized fits the calculation applies to runs satisfying the round conditions.

6. Interpret the result

1. How many successful rounds suffice?

For a training-error target $\eta \in (0, 1)$ and common lower edge γ, a sufficient integer budget is

$$T \geq \left\lceil \frac{\log\left(\frac{1}{\eta}\right)}{2\gamma^2} \right\rceil.$$

At $\eta = 0.01$, a lower edge of 0.10 gives 231 rounds. Raising the lower edge to 0.20 gives 58. These budgets depend on the edge holding throughout. Actual training error can reach the target earlier.

2. What exactly did the round minimize?

In $F = \sum_j \beta_j h_j$, coordinate j is the coefficient of base function h_j. For the selected function h, with weighted error ε,

$$\left[\frac{d}{da} L(F + ah)\right]_{a=0} = -L(F)(1 - 2\varepsilon).$$

Positive edge gives descent for a small positive step. The AdaBoost coefficient minimizes loss along this chosen direction. Certifying the steepest available direction also requires a selection rule.

3. Choosing a stopping round

Even after every training label is correct, signed scores and exponential loss can change. Fit candidate rounds on training data and select a stopping round using validation loss. Evaluate the completed choice on separate test data or with outer cross-validation. The training bound alone cannot select a round for predicting new labels.